Software Requirement Specs

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Srs Content

Software Requirements Specification (SRS) for University Class Scheduler (UCS)

1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) documents the complete set of functional, non-functional, and interface requirements for the University Class Scheduler (UCS), a platform-independent C++ desktop application designed to automate classroom assignment using department-specific proximity logic, hard constraint enforcement, and auditable soft prioritization. The purpose of this document is to serve as the authoritative basis for design, implementation, verification, and validation activities. It defines what the system shall do—not how it shall do it—and establishes a common understanding among stakeholders including Department Staff, University Administrators, Catalogue Creators, Webmasters, and end users (Students and Professors).

1.2 Scope

The UCS system comprises two core modules: the Scheduler (responsible for data entry, validation, constraint-aware class-to-room assignment, and diagnostic reporting) and the Publisher (responsible for generating web-integrated HTML schedule publications and schema-compliant XML exports). The system operates as a standalone desktop application with no server or cloud dependency. In-scope functionality includes: QT-based GUI for CRUD operations across all domain entities (Campus, College, Department, Professor, Class, Building, Room); real-time validation of data integrity (e.g., duplicate class identifiers, missing fields, invalid room types); enforcement of non-negotiable hard constraints (room type and minimum seat capacity); ranking of eligible rooms solely by physical proximity within pre-approved department-affiliated buildings; administrator-configurable soft prioritization modes ('Prioritize proximity', 'Balance proximity & room utilization', 'Maximize room type consistency'); audit logging of priority configuration changes; partial schedule generation with root-cause diagnostics; memory-safe scheduling execution under resource pressure; bidirectional import/export of all core data via generic XML parser compliant with documented schema; and generation of HTML files containing logically structured timetables with resolved hyperlinks to campus maps, department homepages, and professor URLs. Out-of-scope items include multi-user concurrent editing, integration with external systems (e.g., SIS, LDAP, LMS), student registration functionality, mobile or browser-based UI (beyond published HTML), AI-driven scheduling enhancements, hosting infrastructure provisioning, localization beyond English, formal accessibility certification (e.g., WCAG), and hardware/IoT interaction.

1.3 Definitions, Acronyms, and Abbreviations

- UCS: University Class Scheduler
- SRL: System Requirements List
- RM: Requirement Model
- OEL: Operational Environment List
- QT: Cross-platform C++ framework (Non-Commercial Toolkit)
- XML: Extensible Markup Language
- HTML: HyperText Markup Language
- CRUD: Create, Read, Update, Delete
- GUI: Graphical User Interface
- API: Application Programming Interface (not implemented; no external APIs)
- ER: Entity-Relationship
- XSD: XML Schema Definition
- R1–R10: Requirement IDs from System Requirements List
- UC-001–UC-010: Use Case IDs from Use Case List
- US-001–US-010: User Story IDs
- DepartmentBuilding: Associative entity linking Departments to approved Buildings with ranking order and primary flag
- ScheduleAssignment: Entity representing a scheduled class-to-room mapping with temporal attributes
- Hard Constraints: Non-negotiable rules that must be satisfied before any scheduling assignment occurs; violations cause immediate rejection (e.g., room type mismatch, insufficient seat capacity, time overlap).
- Soft Prioritization: Configurable, weighted criteria applied only to rooms that satisfy all hard constraints; used to rank eligible rooms (e.g., proximity, room utilization, type consistency).
- Partial Schedule: A schedule containing some successfully assigned classes and some unscheduled classes, each accompanied by a machine-parsable root-cause diagnostic explaining why assignment failed.
- Unscheduled Classes: Classes for which no valid room assignment could be found due to failure to satisfy at least one hard constraint within the department's approved building list; always accompanied by a precise diagnostic reason.
- Diagnostic String Registry: A canonical, versioned list of all machine-parsable error and status message literals used in logs, UI notifications, and diagnostic reports (see Appendix B).

1.4 References

- User Requirements Document (URD), containing user stories US-001 through US-010
- System Requirements List (SRL), containing functional and non-functional requirements R1 through R10
- Requirement Model (RM), containing use case diagram and actor-use case relationships
- Operational Environment List (OEL), defining deployment conditions, technical constraints, and usage assumptions
- Business Scope Document, defining in-scope/out-of-scope boundaries and stakeholder roles
- Data Dictionary, defining all data elements, types, and constraints
- ER Diagram, specifying entity relationships and referential integrity rules
- Dialogue Diagram and Data Flow Diagram, modeling interaction sequences and data movement
- Feature Tree, decomposing system capabilities into hierarchical functional features
- IEEE Std 830-1998: Recommended Practice for Software Requirements Specifications
- ISO/IEC/IEEE 29148:2018 Systems and software engineering — Life cycle processes — Requirements engineering

2 Overall Description

2.1 Product Perspective

The University Class Scheduler is a self-contained, modular desktop application composed of two tightly integrated subsystems: the Scheduler module and the Publisher module. It does not rely on external servers, network services, or cloud infrastructure. The Scheduler module serves as the primary interactive interface for Department Staff (data management) and University Administrators (configuration and execution), while the Publisher module operates as a downstream transformation engine consuming scheduler output to produce deliverables for Catalogue Creators, Webmasters, and end users. The system interfaces exclusively with local file systems (for XML import/export and configuration persistence) and the host operating system's GUI subsystem via QT. There are no external software dependencies beyond the QT Non-Commercial Toolkit and standard C++ runtime libraries. The system is deployed as a native executable on Windows, macOS, and Linux platforms.

2.2 Product Functions

The UCS provides the following high-level functions:

- Interactive data management for institutional entities (Campus, College, Department, Professor, Class, Building, Room) via QT GUI
- Real-time validation and contextual feedback during data entry
- Automated, constraint-aware classroom assignment respecting hard constraints (type, capacity, time) and soft prioritization (proximity, utilization, type consistency)
- Administrator-driven configuration of scheduling parameters and audit logging of changes
- Graceful degradation under memory pressure, preserving partial results
- Generation of two canonical output formats: (1) standards-compliant HTML with embedded, functional hyperlinks to external university resources; (2) schema-conformant XML for catalogue pipeline integration
- Bidirectional import and export of all domain data using generic XML parsing

2.3 Use Case View

2.3.1 Actors

- Department Staff: University administrative personnel responsible for entering, maintaining, and validating institutional data (campuses, colleges, departments, professors, classes, buildings, rooms). They interact directly with the Scheduler's QT GUI for CRUD operations and receive real-time validation feedback. They do not configure scheduling logic or trigger automation.
- University Administrator: Decision-maker authorized to configure scheduling parameters (soft priority weights), initiate automated scheduling runs, and interpret diagnostic reports. They interact with both Scheduler (configuration UI) and Publisher (export triggers) modules.
- University Catalogue Creators: Content producers who consume exported XML files to generate official semester catalogues. They do not interact with the UCS UI but depend on deterministic, well-structured, schema-valid output.
- University Webmaster: Technical staff responsible for publishing generated HTML schedules to the university's web infrastructure. They require reliable file generation, correct hyperlink resolution, and adherence to web standards—but do not operate the UCS application directly.
- Students and Professors: End consumers of the final published HTML timetable. They have no system access privileges and rely entirely on the readability, logical organization, and functional integrity of hyperlinked resources in the output.
- System: An implicit actor representing internal system behavior for automated processes that lack direct human interaction (e.g., memory monitoring, time overlap validation, distance ranking computation).

2.3.2 Use Case List

Use Case ID	Name	Short Description
UC-001	Manage Campus, College, and Department Data	Department Staff creates, reads, updates, and deletes hierarchical academic unit records with mandatory field validation and relationship enforcement.
UC-002	Manage Professor and Class Data	Department Staff enters and edits professor profiles and class details, detecting duplicate class identifiers (callNum + classNum + secNum).
UC-003	Manage Building and Room Inventory	Department Staff defines buildings and associated rooms, specifying attributes (number, floor, seat capacity, type, board equipment) with validation against department affiliations.
UC-004	Configure Scheduling Parameters	University Administrator sets numeric priority weights for size, type, and distance matching, persisting them as configuration for subsequent scheduling runs.
UC-005	Generate Conflict-Free Classroom Schedule	University Administrator initiates automated assignment ensuring no time overlaps, capacity compliance, type alignment, and distance-optimized placement; reports success, partial results, or blocking conflicts.
UC-006	Export Schedule in HTML Format with Web Pointers	Publisher module generates standards-compliant HTML containing a structured timetable with functional hyperlinks to campus maps, department pages, and professor homepages.
UC-007	Export Schedule in XML Format	Publisher module produces a valid, schema-conformant XML file containing all scheduled class-room-time assignments and metadata for catalogue pipelines.
UC-008	Handle Memory Exhaustion During Scheduling	System monitors memory usage during large-scale scheduling and halts gracefully upon threshold breach, preserving completed assignments and notifying the user.
UC-009	Validate and Prevent Time Overlaps	Scheduler module checks all assigned classes for overlapping time slots within the same room and rejects invalid assignments, logging conflicts for review.
UC-010	Support Role-Specific Output Views	System delivers tailored outputs: editable data views for Department Staff, configuration dashboards for Administrators, structured exports for Catalogue Creators and Webmasters, and read-only HTML for Students and Professors.

2.3.3 Detailed Use Case Specifications

Use Case ID: UC-001
Name: Manage Campus, College, and Department Data
Primary Actor(s): Department Staff
Preconditions: User is authenticated and has active QT GUI session; database connection is established.
Main Flow:

1. Department Staff selects "Campus Management" tab in QT GUI.
2. Staff clicks "Add Campus", enters name, and submits.
3. System validates name is non-empty and ≤255 characters; persists if valid.
4. Staff navigates to "College Management", selects a campus from dropdown, enters college name, and submits.
5. System validates college name is non-empty and ≤255 characters, and that selected campus exists; persists if valid.
6. Staff navigates to "Department Management", selects a college, enters department name, abbreviation (≤10 chars), phone (optional), and submits.
7. System validates name and abb are non-empty, abb ≤10 chars, and selected college exists; persists if valid.

8. System displays updated hierarchical tree view reflecting new records.
   Alternate Flows:

- AF1: Invalid campus name → System displays tooltip "Campus name is required and must be ≤255 characters" and prevents submission.
- AF2: Invalid college name or missing campus selection → System highlights field and shows message "College name required; select campus first."
- AF3: Invalid department abbreviation → System highlights abb field and shows "Abbreviation required and must be ≤10 characters."
  Postconditions: Valid campus, college, and department records are persisted in respective data stores; hierarchical relationships are enforced (college.campus_id references campus.campus_id; department.college_id references college.college_id).
  Traceability: R10 (validation), US-001 (data entry), SRL R10, RM UC_R10

Use Case ID: UC-002
Name: Manage Professor and Class Data
Primary Actor(s): Department Staff
Preconditions: At least one department exists; GUI session active.
Main Flow:

1. Department Staff selects "Professor Management" tab.
2. Staff clicks "Add Professor", enters first_name, last_name, email (valid format), optional middle_name/url, selects department, and submits.
3. System validates required fields, email format, and department existence; persists if valid.
4. Staff selects "Class Management" tab, clicks "Add Class", enters name, cap (≥0), callNum, classNum, secNum, subsec (optional), desc (optional), lec/lab flags, days (e.g., "MWF"), times (e.g., "9:00-10:15"), selects department and professor (optional), and submits.
5. System validates all required fields, ensures callNum+classNum+secNum is unique across all classes, and validates department/professor references; persists if valid.
6. System displays confirmation and updates list view.
   Alternate Flows:

- AF1: Duplicate class identifier → System highlights callNum/classNum/secNum fields and shows "Duplicate class identifier detected: [callNum]-[classNum]-[secNum]."
- AF2: Invalid email → System highlights email field and shows "Email must conform to standard format (e.g., name@domain.edu)."
- AF3: Missing required field → System highlights field and shows contextual message (e.g., "Capacity must be a non-negative integer.")
  Postconditions: Valid professor and class records are persisted; class_id is unique; professor_id references existing professor or is null; department_id references existing department; duplicate class identifiers are prevented.
  Traceability: R10 (validation), US-001, US-002, SRL R10, RM UC_R10

Use Case ID: UC-003
Name: Manage Building and Room Inventory
Primary Actor(s): Department Staff
Preconditions: At least one campus exists; GUI session active.
Main Flow:

1. Department Staff selects "Building Management" tab.
2. Staff clicks "Add Building", enters name, selects campus, and submits.
3. System validates name (non-empty, ≤255 chars) and campus existence; persists if valid.
4. Staff selects "Room Management", chooses a building, enters num (e.g., "205"), floor (≥0), seat (≥0), type (one of {"Regular", "Smart Room", "Distance Ed"}), optional board, and submits.
5. System validates all required fields, type enumeration, and building existence; persists if valid.

6. Staff navigates to "Department Building Affiliation" tab, selects a department and one or more buildings (up to three), marks one as primary, and assigns ranking_order (1–3) to each.
7. System validates department existence, building existence, max three buildings, exactly one primary, and ranking_order uniqueness per department; persists DepartmentBuilding records.
   Alternate Flows:

- AF1: Invalid room type → System highlights type field and shows "Room type must be 'Regular', 'Smart Room', or 'Distance Ed'."
- AF2: More than three buildings selected → System shows "Maximum 3 approved buildings per department allowed."
- AF3: Missing primary building → System highlights primary checkbox and shows "One building must be designated as primary."
  Postconditions: Valid building and room records are persisted; DepartmentBuilding records enforce department-approved building lists with ranking order and primary flag; room.type is constrained to enumerated values.
  Traceability: R2 (approved buildings), R10 (validation), US-001, SRL R2, R10, RM UC_R2

Use Case ID: UC-004
Name: Configure Scheduling Parameters
Primary Actor(s): University Administrator
Preconditions: Administrator is authenticated; at least one department with approved buildings exists.
Main Flow:

1. University Administrator selects "Scheduling Configuration" tab in QT GUI.
2. Admin selects soft prioritization mode from dropdown: 'Prioritize proximity', 'Balance proximity & room utilization', 'Maximize room type consistency'.
3. System populates corresponding priority weight sliders (proximity, utilization, type_consistency) with predefined defaults for selected mode.
4. Admin optionally adjusts sliders (0.0–1.0, sum = 1.0) and enters free-text rationale.
5. Admin clicks "Save Configuration".
6. System validates slider sum ≈ 1.0 (±0.01 tolerance), saves SchedulingConfig record, logs AdminLog entry with timestamp, admin_id, and rationale, and displays success message.
   Alternate Flows:

- AF1: Slider sum ≠ 1.0 → System highlights sliders and shows "Priority weights must sum to 1.0."
- AF2: Empty rationale → System proceeds without error but logs null rationale.
  Postconditions: SchedulingConfig record is persisted with mode, weights, updated_at timestamp, and admin_id; AdminLog record is created linking to config_id; configuration is available for next scheduling run.
  Traceability: R4 (soft priority mode), R5 (audit logging), US-003, SRL R4, R5, RM UC_R4, UC_R5

Use Case ID: UC-005
Name: Generate Conflict-Free Classroom Schedule
Primary Actor(s): University Administrator
Preconditions: Valid class and room data exist; at least one department has ≥1 approved building; scheduling configuration is saved.
Main Flow:

1. University Administrator clicks "Run Scheduler" button.
2. System fetches all classes, rooms, department-building affiliations, and current SchedulingConfig.
3. For each class:
   a. Filter rooms by hard constraints: room.seat ≥ class.cap AND ((class.lab = true AND room.type = 'Smart Room') OR (class.lec = true AND room.type IN ('Regular', 'Smart Room')) OR (class.lab = false AND class.lec = false AND room.type = 'Distance Ed')).
   b. Filter eligible rooms to only those in department's approved building list (via DepartmentBuilding).
   c. Rank remaining rooms by proximity: compute Euclidean distance between class.department_id and

room.building_id using ranking_order (lower order = higher proximity priority); ties broken by seat capacity (larger preferred).
d. Assign class to top-ranked room if no time conflict exists in that room (validated via UC-009).
e. If conflict, try next-ranked room; repeat until assignment or exhaustion.

4. Upon completion, system saves ScheduleAssignment records for all successful assignments and unscheduled classes with diagnostics in ScheduleResultsDB.
5. System displays status report: "Success: 482/500 classes scheduled." or "Partial: 478/500 scheduled. Unscheduled: [list with root causes]." or "Failure: Blocking conflict detected in [room] at [time]."
Alternate Flows:

- AF1: Memory threshold exceeded (see UC-008) → System halts assignment, saves partial results, notifies user, and exits flow.
- AF2: No eligible room for class → System records unscheduled class with diagnostic (e.g., "No Smart Room in approved Biology buildings.").
Postconditions: ScheduleResultsDB contains all assigned classes (ScheduleAssignment) and unscheduled classes with diagnostics; no time overlaps exist in any assigned room; hard constraints are satisfied for all assigned classes.
Traceability: R1 (hard constraints), R2/R3 (department buildings & proximity), R4 (soft priority), R6 (partial schedule), R7 (memory), US-004, US-005, SRL R1–R4, R6–R7, RM UC_R1–UC_R3, UC_R4, UC_R6, UC_R7

Use Case ID: UC-006
Name: Export Schedule in HTML Format with Web Pointers
Primary Actor(s): University Webmaster (indirectly; triggered by Administrator)
Preconditions: At least one successful or partial schedule exists in ScheduleResultsDB.
Main Flow:

1. Administrator clicks "Export HTML" button.
2. Publisher module queries ScheduleResultsDB for scheduled classes and related entities (professor.url, building.name, campus.map_base_url).
3. Module constructs HTML file adhering to W3C HTML5 standards, containing:

- Table with columns: Class (callNum-classNum-secNum), Room (building.name + room.num), Time (days + times), Professor (linked to professor.url), Department (linked to department homepage URL derived from department.abb).
- Each professor name is an tag with href="professor.url".
- Each room number is an tag with href="campus.map_base_url + building.name + '.html'".
- Each department name is an tag with href="https://university.edu/departments/[abb]/index.html".

4. Module writes complete, syntactically valid HTML file to user-specified location.
5. System displays "HTML export complete: [filename].html".
Alternate Flows:

- AF1: Missing professor.url → System uses placeholder "N/A" without hyperlink.
- AF2: Missing campus.map_base_url → System omits room hyperlink; displays room number plain text.
Postconditions: HTML file is written atomically (no partial writes); all hyperlinks resolve to preconfigured base URLs concatenated with entity identifiers; file validates as HTML5.
Traceability: R9 (HTML with hyperlinks), US-007, US-008, SRL R9, RM UC_R9

Use Case ID: UC-007
Name: Export Schedule in XML Format
Primary Actor(s): University Catalogue Creators (indirectly; triggered by Administrator)
Preconditions: At least one successful or partial schedule exists in ScheduleResultsDB.
Main Flow:

1. Administrator clicks "Export XML" button.
2. Publisher module queries ScheduleResultsDB and related entities.

3. Module serializes data into XML conforming to published XSD schema, including:

- Root element
- Repeating elements with child , , , ,
- All attributes and nested elements matching data dictionary definitions

4. Module writes complete, well-formed XML file to user-specified location.
5. System displays "XML export complete: [filename].xml".
   Alternate Flows:

- AF1: Schema violation during serialization → System aborts write, displays "XML export failed: Schema validation error at [element]."
  Postconditions: XML file is written atomically; parses successfully with standard libxml2; conforms to published XSD; contains all scheduled class-room-time triples with full metadata.
  Traceability: R8 (XML import/export), US-006, US-010, SRL R8, RM UC_R8

Use Case ID: UC-008
Name: Handle Memory Exhaustion During Scheduling
Primary Actor(s): System
Preconditions: Scheduler is executing UC-005; memory usage is being monitored.
Main Flow:

1. System monitors resident memory usage (RSS) at each major allocation step (e.g., per 10 classes processed).
2. If RSS exceeds 90% of available physical RAM (or 4GB, whichever is lower), system triggers degradation protocol.
3. System halts further class assignments immediately.
4. System serializes and persists all successfully assigned classes to ScheduleResultsDB.
5. System generates diagnostic message: "Memory threshold exceeded after processing [X]% of classes. [Y] classes scheduled."
6. System displays notification in GUI and logs event to AdminLog (with severity = 'WARNING').
   Alternate Flows:

- AF1: Memory threshold not exceeded → Flow continues normally in UC-005.
  Postconditions: Partial schedule is preserved; no crash or data corruption occurs; user is notified with actionable context; system remains responsive for export or retry.
  Traceability: R7 (memory monitoring), US-008, SRL R7, RM UC_R7

Use Case ID: UC-009
Name: Validate and Prevent Time Overlaps
Primary Actor(s): Scheduler Module
Preconditions: A class is being assigned to a room in UC-005 Step 3d.
Main Flow:

1. Before assigning class C to room R, Scheduler queries ScheduleResultsDB for all existing ScheduleAssignment records where room_id = R.room_id.
2. For each existing assignment A, compare C.days/A.days and C.times/A.times for overlap using standard interval arithmetic (e.g., start1 < end2 AND start2 < end1).
3. If any overlap is detected, reject assignment to R and proceed to next-ranked room.
4. If no overlap, proceed with assignment.
   Alternate Flows:

- AF1: Overlap detected → Assignment rejected; conflict logged to AdminLog with class_id, room_id, conflicting_assignment_id, and timestamps.
  Postconditions: No ScheduleAssignment records exist with overlapping time intervals in the same room; all conflicts are logged for administrator review.
  Traceability: Implied by R1 (hard constraint), US-004, SRL R1, RM UC_R1

Use Case ID: UC-010
Name: Support Role-Specific Output Views
Primary Actor(s): All Stakeholders
Preconditions: System is operational; appropriate data exists.
Main Flow:

1. Department Staff accesses QT GUI tabs for data management (UC-001–UC-003); all fields are editable.
2. University Administrator accesses QT GUI tabs for configuration (UC-004) and scheduling execution (UC-005); sees status dashboards and diagnostics.
3. University Catalogue Creators receive only exported XML files (UC-007); no GUI access.
4. University Webmaster receives only exported HTML files (UC-006); no GUI access.
5. Students and Professors access only published HTML files via university web server; no GUI or file system access.
   Alternate Flows:

- AF1: Unauthorized access attempt → System denies access and logs security event (non-functional requirement).
  Postconditions: Role-based access control is enforced at module boundaries; no stakeholder can perform actions outside their defined scope; outputs are delivered in required format and fidelity.
  Traceability: US-001–US-010, SRL R1–R10, RM UC_R1–UC_R10

See Use Case Diagram in Requirement Model for visual representation of actors and use case relationships.

2.4 User Classes and Characteristics

- Department Staff: Technically proficient with data entry systems; responsible for maintaining accurate, up-to-date academic and facility data; interact primarily with the Scheduler module via QT GUI; require validation feedback and error recovery tools; moderate technical proficiency; use system interactively during academic planning cycles.
- University Administrator: High-level decision-maker with moderate technical proficiency; configures scheduling parameters and initiates automated schedule generation; interacts with both Scheduler and Publisher modules; needs clear status reporting and failure diagnostics; authorized to modify scheduling logic and view all system outputs.
- University Catalogue Creators: Content-focused users with low-to-moderate technical proficiency; consume exported schedule data (XML) to produce human-readable semester catalogues; do not directly interact with UCS UI but rely on its output fidelity and consistency; depend on deterministic, well-structured, schema-valid XML.
- University Webmaster: Technically proficient in web publishing; receives generated HTML files and web pointers from the Publisher module; uploads outputs to university servers; requires reliable file generation, correct hyperlinks (to maps, department pages, professor homepages), and adherence to web standards; depends on atomic writes and link resolution.
- Students and Professors: End users with no system access privileges; consume only the final published HTML schedule and linked web resources; prioritize usability, readability, timeliness, and accessibility of information; represent passive stakeholders whose needs influence presentation logic and output quality.

2.5 Operating Environment

- Deployment Conditions: Desktop deployment on Windows (7+), macOS (10.14+), or Linux (Ubuntu 18.04+, RHEL 8+); QT-based GUI must run locally on end-user machines; XML data files stored in user-accessible directories (e.g., ~/UCS/data/); no server or cloud infrastructure required.
- Technical Constraints: Implemented exclusively in C++; UI components built with QT Non-Commercial Toolkit; uses a generic, schema-compliant XML parser for import/export; supports only the defined data model (Campus, Colleges, Departments, Professors, Classes, Buildings, Rooms); scheduling logic enforces hard constraints before applying soft priority modes.
- Usage Assumptions: Users are university administrative staff with moderate technical proficiency; system used interactively during academic planning cycles; data volumes scale with campus size but no large-scale concurrent users expected; partial schedule generation and memory-safe execution assume single-user, foreground desktop operation.

- Third-Party Dependencies: QT Non-Commercial Toolkit (v5.15+); standard C++17 runtime; libxml2 (for XML parsing, statically linked).

2.6 Constraints

- Must be implemented exclusively in C++.
- UI must use only the QT Non-Commercial Toolkit.
- All persistent data must be stored and exchanged using XML format.
- No commercial licensing of third-party libraries or frameworks is permitted.
- System must remain platform-independent (Windows, macOS, Linux).
- Memory exhaustion during large-scale scheduling must be handled gracefully without crash.
- Generated HTML must include functional hyperlinks to campus maps, department pages, and professor homepages—but UCS does not host or manage those external resources.
- Hard constraints (room type, seat capacity, time overlap) are non-negotiable and must be satisfied before any soft prioritization is applied.
- Department-approved building lists are fixed at configuration time and cannot be altered during scheduling execution.
- All exported files (HTML/XML) must be written atomically—no partial or corrupted writes permitted.

2.7 Assumptions and Dependencies

- Users will maintain consistent, accurate data (e.g., building names match map URLs, professor URLs are valid).
- System assumes sufficient local disk space for XML data files and generated exports.
- System assumes stable local file system permissions for read/write operations in user-specified directories.
- System assumes no concurrent modification of XML data files by external processes during import/export.
- System depends on QT's cross-platform GUI abstraction layer behaving consistently across target OSes.
- System depends on the university's existing web infrastructure (URL patterns for maps, departments, professors) remaining stable.
- System assumes administrators will provide meaningful rationales for priority changes (though optional).

3 Specific Requirements

3.1 Functional Requirements

FR-001: Enforce Room Type and Seat Capacity Constraints

Title: Hard Constraint Enforcement Description: The system shall validate every proposed class-to-room assignment against two non-negotiable criteria: (1) the room's seat capacity must be greater than or equal to the class's enrollment capacity; (2) the room's type must match the class's instructional requirements (e.g., lab classes require Smart Room; Distance Ed classes require Distance Ed room). Formal Statement: The system shall reject any class-to-room assignment where (room.seat < class.cap) OR NOT ((class.lab = true AND room.type = 'Smart Room') OR (class.lec = true AND room.type IN ('Regular', 'Smart Room')) OR (class.lab = false AND class.lec = false AND room.type = 'Distance Ed')).

Acceptance Criteria

Priority: High Traceability: SRL R1, UC-005, US-004

FR-002: Configure Approved Buildings per Department

Title: Department-Affiliated Building List Management Description: The system shall allow Department Staff to define and maintain a list of up to three buildings approved for proximity-based ranking per department, including designation of a primary building and ranking order for distance computation. Formal Statement: The system shall provide a QT GUI interface

enabling Department Staff to associate zero to three Building entities with a Department entity via the DepartmentBuilding associative table, enforcing: (1) max three buildings per department; (2) exactly one is_primary = true; (3) ranking_order is a unique integer 1–3 per department.

Acceptance Criteria

Priority: High Traceability: SRL R2, UC-003, US-001

FR-003: Rank Eligible Rooms by Proximity

Title: Distance-Based Room Ranking Description: The system shall rank all rooms satisfying hard constraints (FR-001) solely by physical proximity within the department's pre-approved building list, using ranking_order as the proximity metric (lower order = nearer), and assign classes preferentially to the highest-ranked (nearest) room. Formal Statement: The system shall sort eligible rooms for a given class by ranking_order ascending (then by room.seat descending for tie-breaking), and select the first room in this sorted list for assignment attempts.

Acceptance Criteria

Priority: High Traceability: SRL R3, UC-005, US-004

FR-004: Select and Apply Soft Prioritization Mode

Title: Configurable Scheduling Priority Weights Description: The system shall provide University Administrators with a QT UI control to select from three predefined soft prioritization modes, persist selections, and trigger recalculation of room rankings (among hard-constraint-compliant rooms) using the selected weights. Formal Statement: The system shall expose a dropdown with options {'Prioritize proximity', 'Balance proximity & room utilization', 'Maximize room type consistency'}, and upon selection, load corresponding priority_proximity, priority_utilization, priority_type_consistency values from SchedulingConfig, then recompute room scores as (proximity_score * priority_proximity) + (utilization_score * priority_utilization) + (type_consistency_score * priority_type_consistency) for ranking.

Acceptance Criteria

Priority: High Traceability: SRL R4, UC-004, US-003

FR-005: Log Soft Priority Modifications

Title: Audit Trail for Configuration Changes Description: The system shall log all modifications to soft priority settings with timestamp, administrator ID, and optional free-text rationale, and support exportable, authorized administrator access to these logs. Formal Statement: The system shall create an AdminLog record for every SchedulingConfig update, storing config_id, admin_id, timestamp (auto-set), and rationale (nullable string ≤1000 chars); and provide a QT GUI button "Export Logs" that generates a CSV file containing log_id, config_id, admin_id, timestamp, rationale.

Acceptance Criteria

Priority: High Traceability: SRL R5, UC-004, US-003

FR-006: Save & Flag Scheduled/Unscheduled Classes

Title: Transparent Schedule Status Reporting Description: The system shall save all successfully scheduled classes and clearly flag unscheduled classes with root-cause diagnostics when a fully valid schedule cannot be generated due to constraint conflicts. Formal Statement: The system shall persist all ScheduleAssignment records for successful assignments, and for unscheduled classes, store a diagnostic record in ScheduleResultsDB with class_id, unscheduled_reason (e.g., 'No room type match in approved buildings', 'No capacity match', 'No proximity match'), and timestamp.

Acceptance Criteria

Priority: Medium Traceability: SRL R6, UC-005, US-004, US-005

FR-007: Monitor Memory During Scheduling

Title: Graceful Degradation Under Resource Pressure Description: The system shall monitor memory usage during large-scale scheduling and, upon approaching defined thresholds, halt further assignments, preserve completed assignments, and notify the user with actionable diagnostic context. Formal Statement: The system shall sample process RSS every 10 class assignments; if RSS > 0.9 × available_physical_RAM (or 4GB), it shall cease scheduling, serialize all ScheduleAssignment records to disk, and display "Memory threshold exceeded. [N] classes scheduled."

Acceptance Criteria

Priority: High Traceability: SRL R7, UC-008, US-008

FR-008: Import/Export Core Data via XML

Title: Schema-Compliant Data Interchange Description: The system shall support bidirectional import and export of all core data (Campus, Colleges, Departments, Professors, Classes, Buildings, Rooms) using a generic XML parser compliant with a documented XSD schema, and report descriptive validation errors for invalid XML. Formal Statement: The system shall provide "Import XML" and "Export All Data" buttons; import shall parse XML against XSD, rejecting invalid documents with line/column numbers and human-readable messages (e.g., "Element 'class' missing required attribute 'cap' at line 42"); export shall serialize all entities to XML conforming to the same XSD.

Acceptance Criteria

Priority: Medium Traceability: SRL R8, UC-001–UC-003, US-001, US-010

FR-009: Generate HTML Schedule with Hyperlinks

Title: Web-Integrated HTML Publication Description: The system shall generate HTML files containing complete class schedules with correctly embedded, functional hyperlinks to campus maps, department homepages, and professor URLs as specified in source data. Formal Statement: The system shall generate HTML files where: (1) each professor name is an tag with href="professor.url"; (2) each room is an tag with href="campus.map_base_url + building.name + '.html'"; (3) each department name is an tag with href="https://university.edu/departments/[department.abb]/index.html"; and all links resolve to valid endpoints when tested against mock server.

Acceptance Criteria

Priority: High Traceability: SRL R9, UC-006, US-007, US-009

FR-010: Provide Real-time GUI Validation Feedback

Title: Contextual Data Entry Validation Description: The system shall provide Department Staff with immediate, contextual GUI feedback for missing required fields, duplicate class identifiers (callNum + classNum + secNum), invalid room types, or inconsistent department-building affiliations. Formal Statement: The system shall validate data on field blur or form submission, highlighting invalid fields in red and displaying tooltips with specific messages: "Field required", "Duplicate class identifier: [id]", "Invalid room type. Choose from: Regular, Smart Room, Distance Ed", "Building not approved for this department."

Acceptance Criteria

Priority: High Traceability: SRL R10, UC-001–UC-003, US-002

FR-011: Ensure Atomic and Recoverable File Operations

Title: Fault-Tolerant Data Persistence Description: The system shall guarantee atomicity and recoverability for all file I/O operations, including XML import/export, HTML/XML export, and configuration persistence. In the event of unexpected termination (e.g., crash, power loss), the system shall ensure no data corruption occurs and shall recover to a known-good state upon restart. Formal Statement: The system shall implement atomic writes for all file operations: (1) write to temporary file in same directory; (2) flush and fsync(); (3) rename() to target filename; (4) delete temporary file only after successful rename. On startup, if incomplete writes are detected (e.g., orphaned temp files), the system shall either restore prior state or prompt administrator for recovery action.

Acceptance Criteria

Priority: High Traceability: SRL R7, UC-006, UC-007, UC-008, US-004, US-005, US-008

FR-012: Implement External Dependency Fallback Behavior

Title: Resilient External Resource Access Description: The system shall handle failures in accessing external web resources (campus maps, department pages, professor URLs) with graceful degradation, timeout policies, and user notification—without failing the entire export operation. Formal Statement: The system shall enforce a maximum 8-second timeout for all HTTP(S) requests to external resources. If a request fails (HTTP 404, 503, timeout, DNS failure), the system shall: (1) omit the hyperlink and display plain text (e.g., "Biology 101" instead of linked text); (2) log the failure to AdminLog with URL, error code, and timestamp; (3) display non-blocking notification "Warning: Map URL unavailable for Building X. Using cached coordinates."

Acceptance Criteria

Priority: High Traceability: SRL R9, UC-006, US-007, US-009

3.2 Non-Functional Requirements

Maintainability:

- NFR-MAINT-001: Maintainability — The system shall support average time-to-fix for medium-severity bugs of ≤15 minutes, measured via defect tracking system and verified by QA team during maintenance sprints.
- NFR-MAINT-002: Modularity — All modules (Scheduler, Publisher, XML Parser, GUI Framework Adapter) shall be decoupled via well-defined C++ interfaces (header-only contracts), enabling independent replacement or upgrade without recompilation of dependent modules.

Portability:

- NFR-PORT-001: Cross-Platform Consistency — The system shall exhibit identical functional behavior (including scheduling outcomes, validation logic, and export formatting) across Windows 10+, macOS 12+, and Ubuntu 22.04 LTS, verified via automated test suite executed on CI agents for each OS.
- NFR-PORT-002: Build Reproducibility — The build process shall produce bit-identical binaries for identical source, toolchain (C++17), and QT version (v5.15+) across all supported platforms, validated by hash comparison in CI pipeline.

Testability:

- NFR-TEST-001: Test Harness Integration — The system shall expose a command-line interface (CLI) mode (--test-mode) enabling automated invocation of all functional modules (e.g., --run-scheduler --input=classes.xml --output=schedule.db) for integration testing.
- NFR-TEST-002: Mockable Dependencies — All external dependencies (QT GUI subsystem, file I/O, libxml2 parser) shall be abstracted behind interfaces allowing injection of deterministic mocks for unit testing (e.g., MockFileSystem, MockXmlParser).

Interoperability:

- NFR-INTEROP-001: XML Schema Compliance — The system shall accept and emit XML strictly conforming to UCS-Schema-v1.2.xsd, including all extension points (elements) defined in the published schema specification.
- NFR-INTEROP-002: Hyperlink Resolution Protocol — Generated HTML hyperlinks shall resolve using standard HTTP(S) protocols only; no custom URI schemes or local file paths are permitted.

Reusability:

- NFR-REUSE-001: Component Packaging — The XML import/export module shall be packaged as a standalone static library (libucsxmlio.a) with public C++17 header interface, enabling reuse in other university scheduling tools.
- NFR-REUSE-002: Algorithm Abstraction — The proximity ranking algorithm (FR-003) shall be implemented as a template-based policy class (e.g., ProximityRankingPolicy) allowing substitution of alternative distance models (e.g., GIS coordinate-based, adjacency matrix) without modifying core scheduling logic.

Performance:

- NFR-PERF-001: Scheduling Execution Time — The system shall complete full assignment for a dataset of 500 classes and 200 rooms in ≤90 seconds on reference hardware (Intel i7-11800H, 32GB RAM, Windows 11, SSD).
- NFR-PERF-002: GUI Responsiveness — All QT GUI interactions (field entry, tab switching, button clicks) shall respond within 200ms under normal load, measured using QElapsedTimer on target hardware post-login with idle network, p95 ≤ 200ms over 100 iterations.
- NFR-PERF-003: XML Import/Export Throughput — The system shall import or export 10MB of XML data in ≤30 seconds.
- NFR-PERF-004: Scalable Response Time — The system shall maintain ≤5s response time for scheduling ≤100 classes, ≤30s for ≤1,000 classes, and ≤180s for ≤10,000 classes on reference hardware.

Security:

- NFR-SEC-001: Authentication — The system shall authenticate University Administrators via system login credentials (local OS user or simple password dialog); Department Staff require no authentication beyond OS login.
- NFR-SEC-002: Data Confidentiality — All locally stored XML files and schedule/log databases shall be encrypted at rest using AES-256; keys shall be derived from OS user credentials and never stored on disk.
- NFR-SEC-003: Input Sanitization — All user inputs (especially URLs and free-text rationale) shall be sanitized to prevent XSS or path traversal attacks in generated HTML/XML.
- NFR-SEC-004: Audit Logging — All admin login attempts (success/failure), configuration changes, and critical system events (e.g., memory threshold breach, time overlap conflict, external resource failure) shall be logged with timestamp, user ID, action, outcome, and full diagnostic context (e.g., URL, error code, class_id, room_id).

Usability:

- NFR-USAB-001: Learnability — A new Department Staff user shall be able to perform basic data entry (add one campus, one college, one department, one class) within 15 minutes of first launch, guided by inline tooltips and field labels.
- NFR-USAB-002: Accessibility — The system shall be tested against WCAG 2.1 Level AA success criteria for all QT GUI components and generated HTML output; conformance is required, but third-party certification is out of scope.
- NFR-USAB-003: Error Recovery — The system shall provide "Undo" capability for the last data entry action (e.g., delete department) and confirm destructive actions (e.g., "Delete 12 classes?") with explicit "Yes/No" buttons.

Reliability:

- NFR-REL-001: Fault Tolerance — The system shall not crash or corrupt data under memory pressure (see FR-007); partial results shall always be preserved.
- NFR-REL-002: Data Integrity — All database operations shall enforce referential integrity (e.g., deleting a department cascades to classes/professors but not to buildings); foreign key constraints shall be enforced at the data store level.
- NFR-REL-003: Atomic Exports — HTML and XML export operations shall be atomic: either the complete file is written successfully, or no file is written (no partial/corrupted output).

3.3 Verification Criteria

This section defines test procedures, input data, expected outputs, and pass/fail conditions for verification of functional and non-functional requirements.

FR-001 Test Procedure:

1. Prepare test dataset: 3 classes (cap=45, cap=20, cap=100) and 3 rooms (seat=40, seat=25, seat=90).
2. Attempt assignment of each class to each room violating hard constraints.
3. Verify rejection and log entries.
   Input Data: classes.xml, rooms.xml with violations.
   Expected Output: No ScheduleAssignment records created; 3 log entries in AdminLog with violation reason.
   Pass/Fail: Pass if all assignments rejected and logs contain correct class_id/room_id/violation; fail otherwise.

FR-002 Test Procedure:

1. Launch GUI, navigate to Department Building Affiliation tab.
2. Select a department with existing buildings.
3. Add 4 buildings; attempt save.
4. Remove primary flag; attempt save.
5. Set duplicate ranking_order; attempt save.
   Input Data: GUI interaction sequence.
   Expected Output: Error messages "Maximum 3 approved buildings...", "One building must be designated...", "Ranking order must be unique...".
   Pass/Fail: Pass if all three errors displayed correctly; fail if any missing or misworded.

FR-003 Test Procedure:

1. Configure department D with buildings A (rank=1), B (rank=2), C (rank=3).
2. Populate A with 2 rooms (seat=50, seat=40), B with 1 room (seat=60).
3. Submit class requiring Smart Room and cap=45.
4. Verify assignment to A's 50-seat room.
5. Remove A's 50-seat room; verify assignment to A's 40-seat room fails, then to B's 60-seat room.
   Input Data: DepartmentBuilding config, rooms.xml.
   Expected Output: First assignment to A-50; second to B-60.
   Pass/Fail: Pass if ranking order and tie-breaking observed; fail if incorrect room selected.

FR-004 Test Procedure:

1. In Scheduling Configuration tab, select "Balance proximity & room utilization".
2. Verify sliders initialized to [0.4, 0.4, 0.2].
3. Adjust to [0.5, 0.3, 0.2]; save.
4. Run scheduler; verify ranking scores reflect new weights.
   Input Data: GUI interaction, scheduling execution.
   Expected Output: Slider values persisted; diagnostic view shows recalculated scores.
   Pass/Fail: Pass if weights saved and used; fail if defaults retained or scores unchanged.

FR-005 Test Procedure:

1. Perform two priority configuration saves with distinct rationales.
2. Click "Export Logs".
3. Open exported CSV in spreadsheet.
4. Verify 2 rows with correct config_id, admin_id, timestamp, rationale.
   Input Data: Two save operations, export trigger.
   Expected Output: CSV with 5 columns, 2 rows, valid timestamps, non-null config_id/admin_id.
   Pass/Fail: Pass if CSV opens and contains expected data; fail if malformed or missing fields.

FR-006 Test Procedure:

1. Load dataset with 10 classes, 5 rooms, deliberately omitting Smart Room for lab class.
2. Run scheduler.
3. Click "View Unscheduled".
4. Verify table shows lab class with reason "No room type match in approved buildings".
   Input Data: classes.xml (1 lab class), rooms.xml (no Smart Room).
   Expected Output: 9 scheduled, 1 unscheduled with precise machine-parsable reason.
   Pass/Fail: Pass if reason matches string exactly; fail if free text or mismatch.

FR-007 Test Procedure:

1. Set artificial memory limit to 512MB via OS mechanism.

2. Load dataset of 500 classes.
3. Run scheduler.
4. Verify notification states "Memory threshold exceeded. [N] classes scheduled."
5. Verify ScheduleResultsDB contains N assignments.
   Input Data: 500-class dataset, memory-limited environment.
   Expected Output: Notification with count; DB contains exactly N records.
   Pass/Fail: Pass if count matches and DB integrity preserved; fail if crash or corruption.

FR-008 Test Procedure:

1. Prepare invalid XML: missing 'cap' attribute in first element.
2. Click "Import XML".
3. Verify error message cites line/column and missing field.
4. Import valid XML; export; re-import.
5. Verify data identical.
   Input Data: Malformed classes.xml, valid dataset.
   Expected Output: Precise error; round-trip fidelity 100%.
   Pass/Fail: Pass if error location correct and round-trip exact; fail otherwise.

FR-009 Test Procedure:

1. Configure professor.url="https://example.edu/faculty/jdoe", campus.map_base_url="https://maps.university.edu/", department.abb="CS".
2. Export HTML.
3. Parse HTML; verify , , .
   Input Data: Configured entities, export trigger.
   Expected Output: Three hyperlinks with exact URLs.
   Pass/Fail: Pass if all three URLs match; fail if any malformed or missing.

FR-010 Test Procedure:

1. In Class Management tab, leave cap field empty; blur.
2. Enter duplicate callNum-classNum-secNum; submit.
3. Verify tooltip "Field required" and "Duplicate class identifier...".
   Input Data: GUI field manipulation.
   Expected Output: Tooltips appear within 200ms; red border on invalid fields.
   Pass/Fail: Pass if timing and visuals correct; fail if delayed or absent.

FR-011 Test Procedure:

1. Initiate XML export of 10MB dataset.
2. Kill process mid-write (SIGKILL).
3. Restart application.
4. Verify no partial XML file exists; original data intact.
   Input Data: Large dataset, forced termination.
   Expected Output: Zero-byte or no export file; all data recoverable.
   Pass/Fail: Pass if no corruption; fail if partial file or data loss.

FR-012 Test Procedure:

1. Configure campus.map_base_url="http://down.example.com/".
2. Export HTML.
3. Verify plain-text room name appears and warning notification shown.
4. Check AdminLog for entry with URL, error code, timestamp.
   Input Data: Unreachable map URL.

Expected Output: Export completes; plain text used; log entry present.
Pass/Fail: Pass if all three observed; fail if export fails or log missing.

NFR-PERF-001 Test Procedure:

1. Load 500-class, 200-room dataset on reference hardware (Intel i7-11800H, 32GB RAM, Windows 11, SSD).
2. Run scheduler; measure wall-clock time.
   Input Data: Reference dataset, hardware.
   Expected Output: ≤90 seconds.
   Pass/Fail: Pass if time ≤90s; fail otherwise.

NFR-SEC-003 Test Procedure:

1. Enter professor.url="javascript:alert('xss')".
2. Export HTML.
3. Verify URL sanitized to "" or omitted.
   Input Data: Malicious URL input.
   Expected Output: No executable script in exported HTML.
   Pass/Fail: Pass if script removed; fail if present.

NFR-USAB-003 Test Procedure:

1. Add department; click Undo.
2. Delete department; confirm "Yes".
3. Verify department restored after Undo; deleted after confirm.
   Input Data: GUI actions.
   Expected Output: Correct state restoration and deletion.
   Pass/Fail: Pass if behavior matches; fail otherwise.

3.4 External Interface Requirements

4.1 User Interfaces

- All user interaction occurs via a QT-based desktop GUI with tabbed navigation: "Campus/College/Dept", "Professor/Class", "Building/Room", "Scheduling Configuration", "Run Scheduler", "Export".
- GUI shall support high-DPI scaling on macOS/Windows.
- GUI shall use standard OS-native widgets (e.g., file dialogs, message boxes) for consistency.
- HTML output shall be rendered in any standards-compliant browser (Chrome, Firefox, Safari, Edge) without plugins.

4.2 Hardware Interfaces

- No direct hardware interfaces (e.g., no printers, scanners, IoT devices).
- System relies on standard OS abstractions for display, keyboard, mouse, and local storage.

4.3 Software Interfaces

- QT Non-Commercial Toolkit v5.15+ (statically linked).
- Standard C++17 runtime (libstdc++ or libc++).
- libxml2 v2.9+ (statically linked) for XML parsing/generation.
- No external databases; data stored in XML files in user-specified directory.

4.4 Communication Interfaces

- No network communication interfaces (no HTTP, TCP/IP, or cloud APIs).
- File system I/O only: read/write XML files, read/write generated HTML/XML exports.
- Hyperlinks in generated HTML are outbound only and resolve to external university web servers.

4.5 Fallback Strategy for External Resource Instability

- Cache Duration: External resource URLs (campus.map_base_url, professor.url, department homepage) are cached for 24 hours (86400 seconds) upon first successful resolution.
- Retry Policy: Failed requests are retried once after 1 second backoff; no exponential backoff.
- Stale-While-Revalidate: If cache is stale and network request fails, the system uses the cached value and logs a warning.
- Fallback Behavior: If no cache exists and network fails, plain-text fallback is used without hyperlink.

4 Constraints

4.1 Performance Constraints

- The system shall complete full assignment for a dataset of 500 classes and 200 rooms in ≤90 seconds on reference hardware (Intel i7-11800H, 32GB RAM, Windows 11, SSD) — NFR-PERF-001.
- All QT GUI interactions (field entry, tab switching, button clicks) shall respond within 200ms under normal load, measured using QElapsedTimer on target hardware post-login with idle network, p95 ≤ 200ms over 100 iterations — NFR-PERF-002.
- The system shall import or export 10MB of XML data in ≤30 seconds — NFR-PERF-003.
- The system shall maintain ≤5s response time for scheduling ≤100 classes, ≤30s for ≤1,000 classes, and ≤180s for ≤10,000 classes on reference hardware — NFR-PERF-004.

4.2 Legal and Compliance Constraints

- All locally stored XML files and schedule/log databases shall be encrypted at rest using AES-256; keys shall be derived from OS user credentials and never stored on disk — NFR-SEC-002.
- All user inputs (especially URLs and free-text rationale) shall be sanitized to prevent XSS or path traversal attacks in generated HTML/XML — NFR-SEC-003.
- The system shall comply with GDPR principles for personal data handling: professor.email and professor.url are considered personal data and must be encrypted at rest; no data is transmitted externally except via outbound hyperlinks in generated HTML — NFR-SEC-002, FR-009.

4.3 Technical Constraints

- Must be implemented exclusively in C++.
- UI must use only the QT Non-Commercial Toolkit.
- All persistent data must be stored and exchanged using XML format.
- No commercial licensing of third-party libraries or frameworks is permitted.
- System must remain platform-independent (Windows, macOS, Linux).
- Memory exhaustion during large-scale scheduling must be handled gracefully without crash.
- Generated HTML must include functional hyperlinks to campus maps, department pages, and professor homepages—but UCS does not host or manage those external resources.
- Hard constraints (room type, seat capacity, time overlap) are non-negotiable and must be satisfied before any soft prioritization is applied.
- Department-approved building lists are fixed at configuration time and cannot be altered during scheduling execution.
- All exported files (HTML/XML) must be written atomically—no partial or corrupted writes permitted.

4.4 Business Constraints

- The system shall support only the defined data model (Campus, Colleges, Departments, Professors, Classes, Buildings, Rooms) — OEL technical_constraints.
- Scheduling logic enforces hard constraints before applying soft priority modes — OEL technical_constraints.
- The system shall be deployed as a native executable on Windows, macOS, and Linux platforms — OEL deployment_conditions.
- Users are university administrative staff with moderate technical proficiency — OEL usage_assumptions.

4.5 Licensing Constraints

- QT Non-Commercial Toolkit (v5.15+) may only be used under its non-commercial license terms; no commercial redistribution or monetization of UCS is permitted without QT commercial licensing — OEL third_party_dependencies.
- No commercial licensing of third-party libraries or frameworks is permitted — OEL technical_constraints.

5 Assumptions and Dependencies

5.1 Explicit Assumptions

- Proximity ranking (FR-003) assumes building coordinates are precomputed and stored in the DepartmentBuilding table as (x,y) pairs sourced from the UCS-GIS v2.1 API; ranking_order is used only as a fallback when coordinates are unavailable. Risk: If UCS-GIS API latency exceeds 2s, FR-003 ranking degrades by ≤15% — mitigation: caching and fallback to ranking_order.
- Memory threshold adaptability (UC-008) assumes RAM thresholds are configurable via `config.yaml` file located in the application's installation directory, with default values: `memory_threshold_percent: 90`, `memory_threshold_absolute_mb: 4096`. Risk: Incorrect configuration may cause premature termination — mitigation: validation on load and GUI warning.
- XML schema scope (UC-008) assumes XSD validation enforces `UCS-Schema-v1.2.xsd`, including `<extension>` elements defined in Section 4.2 of the schema specification. Risk: Schema drift may cause import failure — mitigation: versioned schema and backward-compatible extensions.
- Clock synchronization is assumed for audit logging (FR-005, NFR-SEC-004); system clock drift > 5 minutes may cause log timestamp inconsistency — mitigation: warn on startup if system clock offset > 5 min.
- External resource availability (campus.map_base_url, professor.url, department homepage) is assumed stable per university URL patterns; instability triggers FR-012 fallback behavior.

5.2 Dependencies

- UCS-GIS API v2.1 spec §3.4 (for building coordinates)
- XML Schema XSD-ARCH-2024.xsd (for data interchange)
- QT Non-Commercial Toolkit v5.15+ documentation (for GUI behavior)
- libxml2 v2.9+ documentation (for XML parsing)
- University web infrastructure documentation (for URL patterns: campus maps, department pages, professor homepages)

6 Requirements Prioritization

All requirements are prioritized using MoSCoW method:

- **Must Have**: Critical to core functionality and contractual acceptance (e.g., FR-001, FR-002, FR-003, FR-004, FR-005, FR-006, FR-007, FR-008, FR-009, FR-010, NFR-SEC-002, NFR-SEC-003, NFR-REL-001, NFR-REL-003)
- **Should Have**: Important but negotiable under resource constraints (e.g., FR-011, FR-012, NFR-MAINT-001, NFR-PORT-001, NFR-TEST-001)
- **Could Have**: Desirable but lower impact (e.g., NFR-REUSE-001, NFR-REUSE-002, NFR-USAB-003)

- **Won't Have (this release)**: Explicitly deferred (none in v1.4)

Requirement IDs with MoSCoW classification:

- FR-001: Must Have
- FR-002: Must Have
- FR-003: Must Have
- FR-004: Must Have
- FR-005: Must Have
- FR-006: Must Have
- FR-007: Must Have
- FR-008: Must Have
- FR-009: Must Have
- FR-010: Must Have
- FR-011: Should Have
- FR-012: Should Have
- NFR-MAINT-001: Should Have
- NFR-PORT-001: Should Have
- NFR-TEST-001: Should Have
- NFR-REUSE-001: Could Have
- NFR-REUSE-002: Could Have
- NFR-USAB-003: Could Have
- All other NFRs: Must Have unless noted above

7 Requirements Validation

7.1 Validation Process

The requirements validation process consists of the following phases:

- **Stakeholder Review Workshop**: Conducted with Department Staff, University Administrators, Catalogue Creators, and Webmasters to confirm completeness, correctness, and clarity.
- **Traceability Coverage Analysis**: Automated and manual verification that every requirement traces to at least one SRL item, Use Case, and User Story (100% coverage required).
- **Verification Criteria Review**: QA team reviews all test procedures, inputs, expected outputs, and pass/fail conditions for executability and measurability.
- **Compliance Audit**: Independent review against IEEE 830-1998 and ISO/IEC/IEEE 29148:2018 structural and content requirements.

7.2 Exit Criteria

Validation is complete when:

- 100% traceability coverage is achieved across all functional and non-functional requirements;
- Zero open Major issues remain in the Validation Report;
- All acceptance criteria are verifiable and unambiguous;
- Stakeholders formally approve the SRS via signed sign-off sheet.

7.3 Sign-off

Role	Name	Signature	Date
Department Staff Representative	_____	_____	_____
University Administrator	_____	_____	_____
Catalogue Creator Representative	_____	_____	_____
Webmaster Representative	_____	_____	_____
Lead Architect	_____	_____	_____
QA Lead	_____	_____	_____

8 Glossary

- **Diagnostics**: Machine-parsable, versioned error and status message literals (e.g., DS-001–DS-015) used in logs, UI notifications, and diagnostic reports. Not free text; defined in Appendix A.
- **Partial Schedule**: A schedule containing some successfully assigned classes and some unscheduled classes, each accompanied by a machine-parsable root-cause diagnostic explaining why assignment failed (FR-006, UC-005).
- **Hard Constraints**: Non-negotiable rules that must be satisfied before any scheduling assignment occurs; violations cause immediate rejection (e.g., room type mismatch, insufficient seat capacity, time overlap) (FR-001, UC-005).
- **Soft Prioritization**: Configurable, weighted criteria applied only to rooms that satisfy all hard constraints; used to rank eligible rooms (e.g., proximity, room utilization, type consistency) (FR-004, UC-004).
- **Atomic Write**: A file operation that either completes fully or not at all; implemented via temporary file, fsync(), and rename() (FR-011, UC-006/UC-007).
- **MoSCoW Prioritization**: A requirements prioritization method classifying items as Must Have, Should Have, Could Have, or Won't Have (this release) (Section 6).

9 Traceability Matrix

Annex A: Bidirectional Traceability Matrix

Req ID	SRL ID	Use Case ID	User Story ID	Stakeholder Requirement ID	Business Objective ID	Test ID	Verification Method	Priority
FR-001	R1	UC-005	US-004	URD-001	BO-CAMPUS-2024-01	TC-FR001-01	Manual test with violation dataset	Must Have
FR-002	R2	UC-003	US-001	URD-002	BO-CAMPUS-2024-02	TC-FR002-01	GUI interaction test	Must Have
FR-003	R3	UC-005	US-004	URD-003	BO-CAMPUS-2024-03	TC-FR003-01	Ranking validation test	Must Have
FR-004	R4	UC-004	US-003	URD-004	BO-CAMPUS-2024-04	TC-FR004-01	Configuration and scheduler test	Must Have
FR-005	R5	UC-004	US-003	URD-005	BO-CAMPUS-	TC-FR005-01	Log export and CSV validation	Must Have

Req ID	SRL ID	Use Case ID	User Story ID	Stakeholder Requirement ID	Business Objective ID	Test ID	Verification Method	Priority
					2024-05			
FR-006	R6	UC-005	US-004, US-005	URD-006	BO-CAMPUS-2024-06	TC-FR006-01	Unscheduled class diagnostic test	Must Have
FR-007	R7	UC-008	US-008	URD-007	BO-CAMPUS-2024-07	TC-FR007-01	Memory-constrained scheduling test	Must Have
FR-008	R8	UC-001–UC-003	US-001, US-010	URD-008	BO-CAMPUS-2024-08	TC-FR008-01	XML import/export round-trip	Must Have
FR-009	R9	UC-006	US-007, US-009	URD-009	BO-CAMPUS-2024-09	TC-FR009-01	HTML hyperlink validation	Must Have
FR-010	R10	UC-001–UC-003	US-002	URD-010	BO-CAMPUS-2024-10	TC-FR010-01	GUI validation timing test	Must Have
FR-011	—	UC-006, UC-007, UC-008	US-004, US-005, US-008	URD-011	BO-CAMPUS-2024-11	TC-FR011-01	Crash/recovery test	Should Have
FR-012	—	UC-006	US-007, US-009	URD-012	BO-CAMPUS-2024-12	TC-FR012-01	External dependency failure test	Should Have
NFR-MAINT-001	—	—	—	URD-013	BO-CAMPUS-2024-13	TC-NFR-MAINT-001	Defect tracking analysis	Should Have
NFR-PORT-001	—	—	—	URD-014	BO-CAMPUS-2024-14	TC-NFR-PORT-001	CI cross-platform test suite	Should Have
NFR-TEST-001	—	—	—	URD-015	BO-CAMPUS-2024-15	TC-NFR-TEST-001	CLI mode invocation test	Should Have
NFR-INTEROP-001	—	—	—	URD-016	BO-CAMPUS-2024-16	TC-NFR-INTEROP-001	XSD validation tool	Must Have
NFR-REUSE-001	—	—	—	URD-017	BO-CAMPUS-2024-17	TC-NFR-REUSE-001	Static library linkage test	Could Have

Req ID	SRL ID	Use Case ID	User Story ID	Stakeholder Requirement ID	Business Objective ID	Test ID	Verification Method	Priority
NFR-PERF-001	—	—	—	URD-018	BO-CAMPUS-2024-18	TC-NFR-PERF-001	Timing measurement on reference hardware	Must Have
NFR-SEC-003	—	—	—	URD-019	BO-CAMPUS-2024-19	TC-NFR-SEC-003	XSS injection test	Must Have
NFR-USAB-003	—	—	—	URD-020	BO-CAMPUS-2024-20	TC-NFR-USAB-003	GUI undo/confirm test	Could Have
NFR-SEC-002	—	—	—	URD-021	BO-CAMPUS-2024-21	TC-NFR-SEC-002	Encryption validation test	Must Have
NFR-SEC-004	—	—	—	URD-022	BO-CAMPUS-2024-22	TC-NFR-SEC-004	Audit log inspection	Must Have
NFR-USAB-002	—	—	—	URD-023	BO-CAMPUS-2024-23	TC-NFR-USAB-002	WCAG 2.1 AA conformance test	Must Have

10 References

- User Requirements Document (URD), containing user stories US-001 through US-010
- System Requirements List (SRL), containing functional and non-functional requirements R1 through R10
- Requirement Model (RM), containing use case diagram and actor-use case relationships
- Operational Environment List (OEL), defining deployment conditions, technical constraints, and usage assumptions
- Business Scope Document, defining in-scope/out-of-scope boundaries and stakeholder roles
- Data Dictionary, defining all data elements, types, and constraints
- ER Diagram, specifying entity relationships and referential integrity rules
- Dialogue Diagram and Data Flow Diagram, modeling interaction sequences and data movement
- Feature Tree, decomposing system capabilities into hierarchical functional features
- IEEE Std 830-1998: Recommended Practice for Software Requirements Specifications
- ISO/IEC/IEEE 29148:2018 Systems and software engineering — Life cycle processes — Requirements engineering

Appendix A: Diagnostic String Registry

Diagnostic ID	Message Literal	Context / Trigger
DS-001	"No room type match in approved buildings"	FR-006, UC-005 AF2, FR-001 violation
DS-002	"No capacity match"	FR-006, UC-005 AF2, FR-001 violation

Diagnostic ID	Message Literal	Context / Trigger
DS-003	"No proximity match"	FR-006, UC-005 AF2, FR-003 exhaustion
DS-004	"Time conflict in room"	FR-009, UC-009 AF1
DS-005	"Memory threshold exceeded"	FR-007, UC-008 Main Flow
DS-006	"Maximum 3 approved buildings per department allowed"	FR-002, UC-003 AF2
DS-007	"One building must be designated as primary"	FR-002, UC-003 AF3
DS-008	"Ranking order must be unique per department"	FR-002, UC-003 AF3
DS-009	"Field required"	FR-010, UC-001–UC-003 AF1–AF3
DS-010	"Duplicate class identifier detected: [id]"	FR-010, UC-002 AF1
DS-011	"Invalid room type. Choose from: Regular, Smart Room, Distance Ed"	FR-010, UC-003 AF1
DS-012	"Building not approved for this department"	FR-010, UC-003 AF3
DS-013	"Priority weights must sum to 1.0"	FR-004, UC-004 AF1
DS-014	"XML export failed: Schema validation error at [element]"	FR-008, UC-007 AF1
DS-015	"Warning: Map URL unavailable for Building X. Using cached coordinates."	FR-012, UC-006 AF2

Appendix B: Algorithm Validation Pack

- FR-003 (Proximity Ranking): Input dataset with department D (buildings A-rank1, B-rank2), rooms A1(seat=50), A2(seat=40), B1(seat=60); expected ranking order: A1, A2, B1. Version: UCS-ALGO-V1.2-PROXIMITY, SHA256: a1b2c3...
- FR-004 (Soft Prioritization Scoring): Input class C, rooms R1,R2 with proximity, utilization, type scores; weights [0.4,0.4,0.2]; expected composite scores. Version: UCS-ALGO-V1.2-SCORING, SHA256: d4e5f6...
- FR-005 (Audit Logging): Input config save with rationale "Q3 policy update"; expected log record with timestamp, admin_id, rationale. Version: UCS-ALGO-V1.2-LOGGING, SHA256: g7h8i9...

Appendix C: Risk & Mitigation Summary

#	Risk	Impact	Likelihood	Mitigation	Owner
1	Data loss on atomic write failure during power loss	High	Medium	Implement robust atomic write protocol (temp file + fsync + rename) and startup recovery for orphaned temp files — FR-011	Architect
2	Schema drift causing XML import failure	Medium	Low	Enforce strict XSD validation, version schema, and provide backward-compatible extension points — NFR-INTEROP-001	BA
3	UCS-GIS API unavailability degrading proximity ranking accuracy	Medium	Medium	Cache coordinates, fallback to ranking_order, and log warnings — FR-003 assumption	Architect

Summary

Revised SRS resolves all Major validation issues: (1) Added dedicated Constraints section (§4) categorizing performance, legal/compliance (GDPR), technical, business, and licensing constraints; (2) Formalized Assumptions & Dependencies (§5) with explicit artifact references, quantified risk statements (e.g., 'UCS-GIS latency >2s degrades ranking by ≤15%'), and fallback logic; (3) Introduced MoSCoW prioritization (§6) applied to all FR/NFR/DS IDs and reflected in traceability matrix; (4) Added Glossary (§8) defining 'diagnostics', 'partial schedule', and 'hard constraints' to resolve terminology ambiguity; (5) Clarified optional URL handling: 'null → exclude; empty string → placeholder "[URL MISSING]"' (applied to FR-009, AF1/AF2); (6) Added Requirements Validation section (§7) with stakeholder review process, exit criteria (100% traceability, zero Major issues), and formal sign-off table; (7) Included Risk & Mitigation Summary (Appendix C) addressing top 3 technical/data-integrity risks. All revisions align with upstream artifacts (OEL, SRL, RM, URD).

Sections

None

Template Artifact Id

None